Check for updates

Received: 17 May 2023 | Revised: 22 April 2024 | Accepted: 23 April 2024

DOI: 10.1111/psyp.14603

PSYCHOPHYSIOLOGY SPR WILEY

EARLY CAREER AWARD PAPER

The moving wave: Applications of the mobile EEG approach to study human attention

Kyle E. Mathewson[1] | **Jonathan P. Kuziek**[2] | **Joanna E. M. Scanlon**[3] | **Daniel Robles**[4]

[1]Department of Psychology, Faculty of Science, University of Alberta, Edmonton, Alberta, Canada

[2]Faculty of Medicine, University of Calgary, Calgary, Alberta, Canada

[3]Department of Psychology, University of Oldenburg, Oldenburg, Germany

[4]Department of Psychology, Rutgers University, Piscataway, New Jersey, USA

Correspondence
Kyle E. Mathewson, Department of Psychology, Faculty of Science, University of Alberta, P-217 Biological Sciences Building, Edmonton, AB T6G 2R3, Canada.
Email: kmathews@ualberta.ca

Abstract

Although historically confined to traditional research laboratories, electroencephalography (EEG) paradigms are now being applied to study a wide array of behaviors, from daily activities to specialized tasks in diverse fields such as sports science, neurorehabilitation, and education. This transition from traditional to real-world mobile research can provide new tools for understanding attentional processes as they occur naturally. Early mobile EEG research has made progress, despite the large size and wired connections. Recent developments in hardware and software have expanded the possibilities of mobile EEG, enabling a broader range of applications. Despite these advancements, limitations influencing mobile EEG remain that must be overcome to achieve adequate reliability and validity. In this review, we first assess the feasibility of mobile paradigms, including electrode selection, artifact correction techniques, and methodological considerations. This review underscores the importance of ecological, construct, and predictive validity in ensuring the trustworthiness and applicability of mobile EEG findings. Second, we explore studies on attention in naturalistic settings, focusing on replicating classic P3 component studies in mobile paradigms like stationary biking in our lab, and activities such as walking, cycling, and dual-tasking outside of the lab. We emphasize how the mobile approach complements traditional laboratory paradigms and the types of insights gained in naturalistic research settings. Third, we discuss promising applications of portable EEG in workplace safety and other areas including road safety, rehabilitation medicine, and brain–computer interfaces. In summary, this review explores the expanding possibilities of mobile EEG while recognizing the existing challenges in fully realizing its potential.

KEYWORDS

attention, cognition, ERP, MoBI, mobile EEG

This is an open access article under the terms of the Creative Commons Attribution-NonCommercial License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited and is not used for commercial purposes.
© 2024 The Authors. *Psychophysiology* published by Wiley Periodicals LLC on behalf of Society for Psychophysiological Research.

1 | INTRODUCTION

For centuries, humans have sought to understand mental processes (Rocca, 2003), but have been limited by the available technology of the times. For example, relative to the early days of psychophysics and experimental psychology, modern tools like wearable and wireless computers and amplifiers used to study mental processes experimentally are very sophisticated. Despite technological advancements, brain imaging still faces challenges, especially in capturing active behaviors like walking and moving (Jungnickel & Gramann, 2016). At the center of these challenges is the ability to measure reliable brain signals of interest while subjects perform complex mobile tasks.

One of the most substantial challenges in mobile EEG research comes from the need to explore naturalistic behaviors (e.g., walking, grasping, etc.) while producing high-quality data despite movements (Makeig et al., 2009). Ensuring this quality is paramount, as it underpins the validity of measures of the cognitive processes we aim to study—whether these processes maintain their integrity outside the lab (construct validity), and whether the EEG findings can predict cognitive or behavioral outcomes in real-world settings (predictive validity).

This review article is structured into three sections. First, we provide an overview of the methodological foundations and the current state of mobile EEG research, summarizing past advancements often involving cumbersome hardware and wires, and describing the challenges and advancements in this field. Next, to showcase one example of the utility of these two techniques, we delve into the specific application of mobile EEG in studying attention within naturalistic settings, showcasing how novel hardware and software in mobile EEG can yield increasingly reliable and meaningful insights into cognitive processes, and underscoring its potential utility. Finally, we explore the broader horizons of mobile EEG, highlighting its relevance and application to diverse domains such as rehabilitation medicine, education, engineering, and social sciences. By exploring the development and application of mobile EEG, we aim to demonstrate its potential to revolutionize cognitive neuroscience research and improve our understanding of brain function and behavior in real-world settings, ultimately paving the way for innovative interventions, treatments, and products.

2 | METHODOLOGICAL FOUNDATIONS OF MOBILE EEG RESEARCH AND FUTURE DIRECTIONS

To lay the groundwork for understanding the applications and future directions of mobile EEG, we must first examine the methodological foundations that underpin this field. In this section, we delve into the methodological foundations that underpin the field of mobile EEG research. We explore the intricacies of electrode selection, portability considerations, and the techniques employed to correct artifacts. We show how low-cost consumer EEG devices are being used to replicate laboratory research in mobile scenarios. We also cast our gaze forward, contemplating future directions and requirements.

2.1 | Challenges in traditional EEG paradigms and the need for mobile approaches?

Building on the methodological foundations outlined above, we now explore the challenges inherent in traditional EEG paradigms and the need for mobile approaches. Electroencephalography (EEG) is a technique that uses scalp electrodes to record brain activity. This technique records the coordinated postsynaptic activity of large clusters of neurons from the cerebral cortex (Olejniczak, 2006). EEG offers extraordinary temporal resolution in the recording of cognitive/attentional processes, as this method can record in the order of milliseconds (St. Louis et al., 2016). Since the development of EEG in the 1920s, scientists have validated many classic paradigms to record brain activity in research laboratories (Cohen, 2017; Shipton, 1975). However, because the magnitude of EEG signals is under 100 microvolts (Parasuraman & Rizzo, 2007), electrical potentials resulting from eyeblinks and body movements can greatly contaminate the recordings (Luck, 2014). To illustrate this, an eyeblink generates noise on the millivolt scale, which is much larger than the microvolt-scale signals typically measured by EEG (Çınar, 2021). Eye movements, including eyeblinks, generate temporary shifts in electric fields that travel around the head surface. These artifactual signals pose major problems for researchers. Completely minimizing artifacts is impossible, and they appear larger than EEG signals of interest recorded during an experiment (Gratton, 1998). The traditional approaches used to deal with artifacts include restricting participant movements during experimental blocks and offline analyses to either discard or correct artifact-related noise (Gwin et al., 2010; Hoffmann & Falkenstein, 2008). Although these solutions increase the data quality of the EEG, conceptually, the isolated and passive environments under which experiments take place make it difficult to generalize findings to everyday situations. For instance, in everyday life, people are bombarded by large amounts of outdoor stimuli and can move freely (Ladouce et al., 2017). The interaction with the environment is one crucial aspect of human

14698986, 2024, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/psyp.14603, Wiley Online Library on [24/08/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

cognition that traditional laboratory approaches are not well suited to explore given the complexity of real-world environments and hardware limitations (Jungnickel & Gramann, 2016). For instance, a conventional laboratory cannot emulate the sensation of riding a bicycle downhill at a very fast speed using an indoor paradigm. It is also impossible to obtain an fMRI recording from someone riding a bicycle downhill outdoors. This limitation highlights the need for novel approaches that can capture cognition in real-world settings, which is precisely the focus of the next section.

2.2 | The interplay of motion, cognition, and ecological validity

In a world where rigorous experimental control has long been considered the standard, why would cognitive neuroscientists want to move away from laboratories into unstructured naturalistic environments? Previous research has shown that cognitive processes are tailored towards an organism's interaction with the physical world (Jungnickel et al., 2018). For example, the concept of *active externalism* proposes that the interaction between a human organism and the external environment creates a coupled system (Clark & Chalmers, 1998). This coupled system is analogous to a cognitive system that influences behaviors and actions. Crucially, the organism–environment interaction is difficult to record in laboratory chambers using conventional recording techniques. Makeig et al. (2009) suggest that cognitive processes have evolved to guide behaviors in the physical environment. They argue that there is a strong link between perception and motor control that facilitates visually guided behavior. For example, environmental information continuously shapes an individual's situational awareness for effective guided action. Also, objects from the environment can be integrated into bodily cognition (a player's racket, e.g.). Moreover, evidence of mirror neurons and self-motion feedback loops during spatial orientation support the argument that brain processes are tightly related to motion control (Gramann et al., 2014).

Studies using dual-task paradigms involving motor control have proposed that there could be shared mechanisms and interactions between areas of cognitive functioning and motor control/sensory processing (Al-Yahya et al., 2011; De Sanctis et al., 2014; Leone et al., 2017). For instance, these studies have explored the degree to which complex tasks can interfere with either cognitive or motor resources necessary for effective task performance. A recent EEG study by (Djebbara et al., 2019) provided evidence that behavioral affordances and action affect how individuals perceive their external environment. In other words, movement influences our perceptual mechanism in an environment. Beyond human literature, animal studies exploring visual neuron activity have shown different activation dynamics in motion relative to idle or immobile states (Gramann et al., 2011). Taken together, these studies show the importance of exploring perceptual, cognitive, and motor processes in dynamic environments to better match the environments under which these processes have evolved.

When it comes to the study of attention during motion, an important implication of the evidence presented above is that there is an interdependence between sensory, motor, and cognitive processing during the execution of behaviors and actions. This interdependence underscores the need for construct validity: ensuring our measures actually assess cognitive processes as they occur in the real world. Since traditional laboratories require subjects to sit still, researchers have questioned the degree to which traditional theories can hold water using more ecologically valid paradigms (Debener et al., 2012). Ecological validity is linked with construct validity and can be thought of as the degree to which real-world and experimental phenomena overlap (Schmuckler, 2001). By considering differences in perceptual experience performing a task outdoors vs. in isolation it becomes clear that mobile paradigms can bring new dimensions to the study of cognitive processes.

In the following two sections, we aim to delineate the capabilities and optimal application contexts of various EEG devices and electrode types, as they transition from consumer to mobile research environments. Our focus is to provide researchers with a comparative understanding of these tools to enable informed decisions based on specific experimental needs, whether in controlled settings or naturalistic scenarios. We will examine the trade-offs between portability, ease of use, and data fidelity, offering guidance on the selection of appropriate technology for mobile EEG paradigms.

2.3 | From consumer EEG devices to mobile EEG paradigms

In the pursuit of validating mobile EEG methodologies, a critical emphasis has been placed on construct validity. Studies demonstrating the detection of P300 components in mobile contexts bolster this, tying back to well-established cognitive theories regarding attentional processes. This link offers a strong foundation for the construct validity of mobile EEG. Predictive validity is equally pivotal, with mobile EEG's capacity to anticipate real-world behavior and outcomes, showing promise in fields like neuroergonomics and rehabilitative medicine. Lastly, face validity is addressed by the consistency of mobile

14698986, 2024, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/psyp.14603, Wiley Online Library on [24/08/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

EEG findings with those from traditional lab-based setups. By replicating findings for classic ERP components like P300 and N2 in dynamic environments, mobile EEG reinforces its credibility and stands as a reliable extension of established neuroscientific research tools.

Given the links between action and cognition, it is important to study perceptual and cognitive processes in paradigms that include motion and naturalistic settings (Lang, 1979). The development of portable, low-cost, and commercially available EEG devices has contributed to the study of cognitive processes under a range of naturalistic behaviors (Xu & Zhong, 2018). These devices greatly reduce the cost of research while allowing for the deployment of paradigms away from the laboratory. One crucial step in the development of mobile paradigms is the replication of laboratory results in less structured settings. For example, the event-related potential component (ERP) known as the P300 or P3, is a positive deflection in amplitude that occurs approximately 300 ms after the onset of a stimulus, and it is associated with modulations in attentional resources applied during stimulus processing and response preparation (Donchin & Coles, 1998; Picton, 1992; Sur & Sinha, 2009). This ERP has been widely replicated in attentional studies due to its robustness and replicability (Mulert et al., 2004). Crucially, consumer EEG systems such as the Emotiv (https://www.emotiv.com/) have been previously modified in walking paradigms to yield high-quality oddball P3 indoors and outdoors and demonstrating that single-trial EEG data quality can be achieved using non-laboratory grade hardware (De Vos et al., 2014; Debener et al., 2012). The Emotiv system has also been deployed to everyday settings such as classrooms to explore students' brain-to-brain synchrony over extended periods (Dikker et al., 2017).

Other commercially available devices such as the meditation tool called Muse Headband (Interaxon Inc.; Figure 1) have been previously used for EEG research. The Muse has been successfully used to record and quantify well-known ERP components such as the N2 and the P3 (Krigolson et al., 2017). In a recent study, our group demonstrated the feasibility of the Muse to predict stroke severity by analyzing different EEG frequency bands using a sample of patients and a portable setup (Wilkinson et al., 2020). We have developed web applications that connect with Bluetooth to Muse EEG systems and allow users to do a hands on tutorial on EEG data collection, processing, and experimental design using their own brain activity (eegedu.com). This online system can be used on mobile phones, affording use in classroom, research, and medical settings. Likewise, work from our workgroup used the Muse as part of a large study to collect EEG signals in remote villages in Malawi (Leal Neto et al., 2021). These results suggest that market-quality EEG tools can provide new alternatives to researchers interested in recording EEG outsite laboratory settings.

Although these commercial devices can pose new challenges (e.g., data quality does not compare to laboratory-grade hardware), their accessibility and ease of use could help to expand ERP research to novel scenarios (Krigolson et al., 2017). Additionally, advancements in portable technology such as miniaturized computers and smartphones have played a key role in the development of mobile paradigms (Lau-Zhu et al., 2019). For example, the Raspberry Pi minicomputer (https://www.raspberrypi.org/) has been used by researchers to successfully record ERP components yielding similar results to traditional laboratory gear (Kuziek et al., 2017). Likewise, the use of tablets (Ladouce et al., 2017) and Android-based smartphones (Bleichner & Debener, 2017; Blum et al., 2017; Debener et al., 2015) for stimulus presentation and data collection represent a significant step towards mobile research for the following reason: Bluetooth based connection reduces the use of cables that constraint subject movement, also, these smaller and lighter devices can be easily deployed to the field (Dehais et al., 2019). This allows for the pursuit of new avenues in field research all over the globe. Although these new technologies make mobile research more accessible and flexible, some scientists have taken a more complex approach to investigate brain/body dynamics under active behaviors.

Over recent years, researchers considered the need for a new approach to record brain dynamics in motion given the views that human behavior is constantly shaped by the environment to maximize behavioral success (Gramann



FIGURE 1 Muse headband showing the four electrode locations (behind each ear and lateral side of forehead, as well as reference and ground electrodes (center three). Adjustable clasps allow a tight fit of the dry electrodes against the head.

14698986, 2024, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/psyp.14603, Wiley Online Library on [24/08/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

et al., 2011). The mobile brain/body imaging (MoBI) approach (Makeig et al., 2009) was developed to collect brain signals, motor behavior, and events occurring within the subject's immediate 3D environment. This approach consists of obtaining synchronized recordings from multiple sources including EEG, artifacts, EMG, movement tracking sensors, and timed events the participant experiences during the experimental paradigm. The challenge of integrating synchronized data required the development of a toolbox for MoBI studies called MoBILAB (Ojeda et al., 2014). This toolbox uses Lab Streaming Layer (LSL) to synchronize the events from the different sensors by estimating the relative event offsets. The different data formats, such as EEG, and EMG are stored separately allowing for the individual processing and visualization of each. The goal of the MoBI approach is to link the relationship between unconstrained behavior, cognitive function, and brain dynamics by analyzing synchronous activity (Jungnickel et al., 2018). This approach radically breaks away from traditional laboratory research where behaviors are generally limited to responses on a computer such as a button pressing.

2.4 | Electrode selection and portability

As we transition into real-world applications of EEG, the selection of electrodes becomes crucial not only for data quality but also for preserving the validity of our findings. The use of wet electrodes, with their lower impedance and higher signal quality, has often been the benchmark for construct validity in laboratory settings. However, their practical limitations in mobile research necessitate the exploration of dry electrodes, which, because their susceptibility to motion artifacts, must be evaluated for their ability to maintain construct validity when recording cognitive processes in naturalistic environments. Moreover, the comparison of ERP signals obtained from these mobile setups to traditional lab-based recordings is essential for establishing face validity, ensuring that the phenomena we observe in the field authentically represent the cognitive processes of interest.

One of the main goals of the mobile/portable EEG field is to transition cognitive neuroscience from highly controlled and restrictive environments into real-world paradigms where subjects behave naturally (Gramann et al., 2014). A key component of this transition is the validation of research paradigms that include portable hardware and data cleaning methods tailored to studying cognition during active behaviors. The following section will present several methods currently used to explore cognition in action followed by a discussion highlighting the strengths and limitations in the field of mobile research.

In mobile research, investigators are faced with many methodological challenges. One of the most crucial ones is producing good-quality data using the EEG sensors or electrodes while subjects are in motion. Ideally, electrodes must be portable enough to allow natural participant movement while being robust enough to remain attached and provide EEG recordings of adequate quality despite that movement (Lau-Zhu et al., 2019). As this mobile field is still in its infancy (Protzak & Gramann, 2018), researchers must determine what type of sensors better suit their research. Common electrode types include wet electrodes that use a conductive gel to enhance conduction or dry electrodes that do not require gel (Leach et al., 2020). Generally, relative to dry electrodes, wet electrodes can achieve lower impedance levels. This greatly enhances the signal quality but one of the major disadvantages of wet electrodes is that they can be time-consuming to set up and require skin abrasion (Hinrichs et al., 2020). On the other hand, dry electrodes can overcome some of the disadvantages of wet electrodes (such as preparation time) even though the signal quality does not measure up to wet sensors (Dehais et al., 2019; Lopez-Gordo et al., 2014).

There are contrasting views regarding the choice of electrodes to use during motion. In a seminal MoBI paper, Makeig et al. (2009) argued in favor of using high density, dry electrodes to prevent gel bridging between adjacent electrodes while allowing for mobility. Due to their flexibility of use (e.g., less time-consuming to set up and no need to re-gel during experimentation), dry electrodes could be more suitable for research involving everyday settings (Ladouce et al., 2019). Moreover, (Gramann et al., 2011) highlighted several issues related to wet electrodes such as the potential for skin allergies and infection if used over a prolonged time. They proposed that dry micro-electromechanical systems, dry fabric electrodes, and hybrid dry electrodes help to overcome these wet-electrode issues. They also noted that relative to wet sensors, dry sensors are not robust enough to tolerate motion artifacts during data collection, likely due to the need for tight mechanical force against the scalp necessary to maintain consistent impedance as opposed to the extra leeway afforded by a barrier of conductive gel (Mathewson et al., 2017). When considering that mobile research prioritizes the collection of signals under naturalistic behaviors this sensitivity to motion artifacts is highly problematic.

In addition to the problem of motion sensitivity, traditional dry sensors have other disadvantages such as not being as comfortable to subjects when used over time (Ladouce et al., 2017), and also when worn tightly which helps maintain good signal quality during experimentation (Mathewson et al., 2017). This can be a potential issue for paradigms where wet electrodes are not suitable for long-term use due to factors such

14698986, 2024, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/psyp.14603, Wiley Online Library on [24/08/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

as the conductive gel becoming dry over time (Dehais et al., 2019). There are, however, new generations of dry electrodes such as the actiCAP Xpress Twist (Brain Products GmbH, Gilching, Germany) which have a wider surface area that may be better tolerated than the traditional dry contact electrodes. Ultimately, due to the aforementioned issues associated with traditional dry electrodes, researchers have proposed other alternatives to collect scalp data during motion.

To enhance the portability of electrodes for mobile studies, a research group has developed an around-the-ear electrode (Figure 2) type called cEEGrid (Bleichner & Debener, 2017; Debener et al., 2015; Mirkovic et al., 2016). Debener et al. (2015) suggested that this multichannel system was designed to overcome many of the issues present in traditional systems: the design is wireless, small, unobtrusive, motion tolerant, and can be tolerated for hours while allowing for accurate recordings of P3 amplitude. Although conceptually cEEGrid electrodes seem to be ideal for mobile research, there are some limitations in this setup. For example, compared to a traditional head cap, these electrodes produce great quality signals in temporal regions but there is considerable signal loss from other scalp regions such as the midline (Meiser et al., 2020). In a recent study, researchers recorded continuous data using the cEEGrid electrodes for 6 hours (Hölle et al., 2021), demonstrating the feasibility of these electrodes for continuous data recording. The authors also reported that all the participants rated the setup as comfortable to wear throughout the day. This has important implications for studies using real-driving situations to measure attention, fatigue, etc.

Previous work from our research group highlights the feasibility of active wet electrodes and flexible/wearable material electrodes for mobile research. By employing a bicycle paradigm (Figure 3), Scanlon has demonstrated that it is possible to collect high-quality EEG signals from subjects during outdoor riding (Scanlon et al., 2020; Scanlon, Townsend, et al., 2019). Furthermore, it has been demonstrated that using active wet electrodes, the overall morphology and topography of the mismatch negativity (MMN) and P3 are similar in subjects completing an oddball task while sitting and during stationary cycling (Scanlon et al., 2017). In a recent study, we compared active and passive EEG setups during standing and walking (Figure 4A; Scanlon et al., 2021). It was found that passive electrode systems can yield similar signal quality relative to an active EEG system as assessed by the signal-to-noise ratio (SNR) and P3 amplitude.

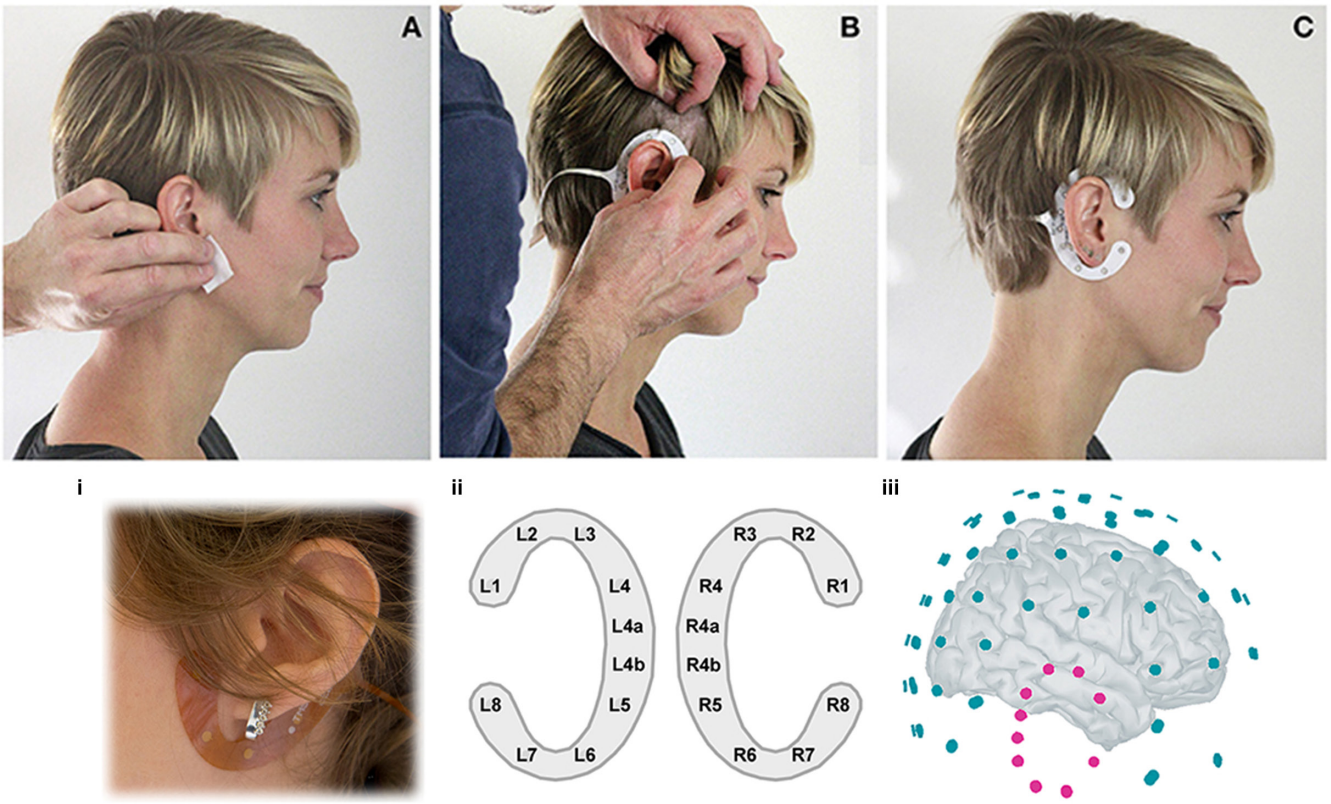


FIGURE 2 cEEGrid Design. Top: Top: Bleichner and Debener (2017). (A) Removal of oils and dirt with alcohol swabs. (B) Adjustment of hair and placement around the ear. (C) Attached device. Bottom: Mirkovic et al. (2016). (i) Unobtrusive device around hair and ear. (ii) Electrode locations and names. (iii) Location in 3D space with respect to the brain and 10–20 locations.

14698986, 2024, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/psyp.14603, Wiley Online Library on [24/08/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

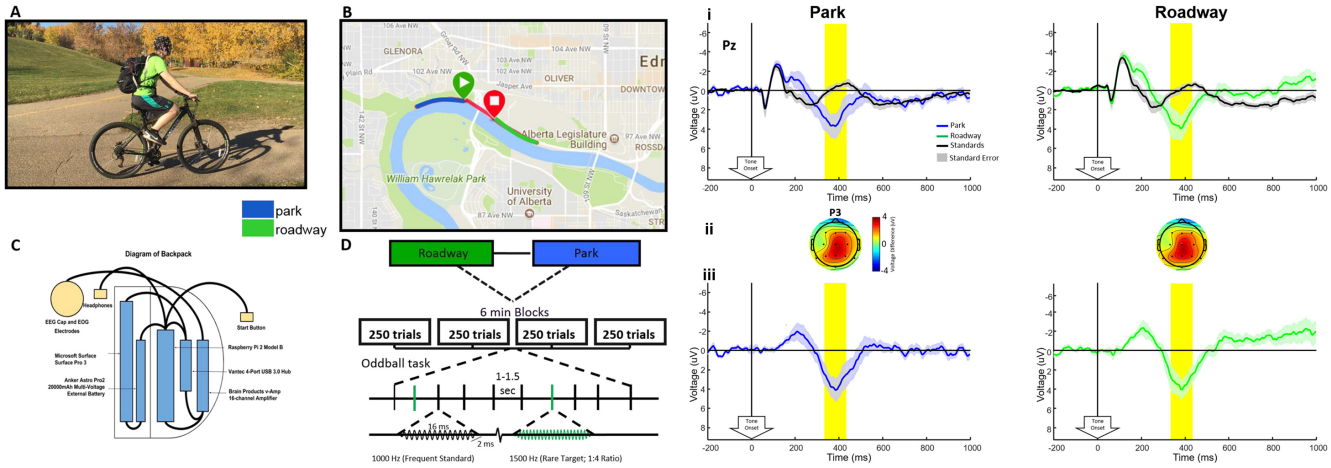


FIGURE 3 Cycling EEG Paradigm. Scanlon et al. (2020). (A) Bike rider on bike near transition between two path conditions in Edmonton, Alberta, Canada. (B) Map showing start and end locations for both condition paths, one along a quiet river and park, the other between the river and a busy roadway. (C) Contents of backpack including portable EEG system and computer. (D) Structure of auditory oddball task done during the ride. Right: (i) Tone elicited ERPs for target and standard tones for the park and roadway conditions. (ii) Topographic maps of the difference between target and standard in the yellow time window. (iii) Difference waves isolating the P3 ERP component in each condition.

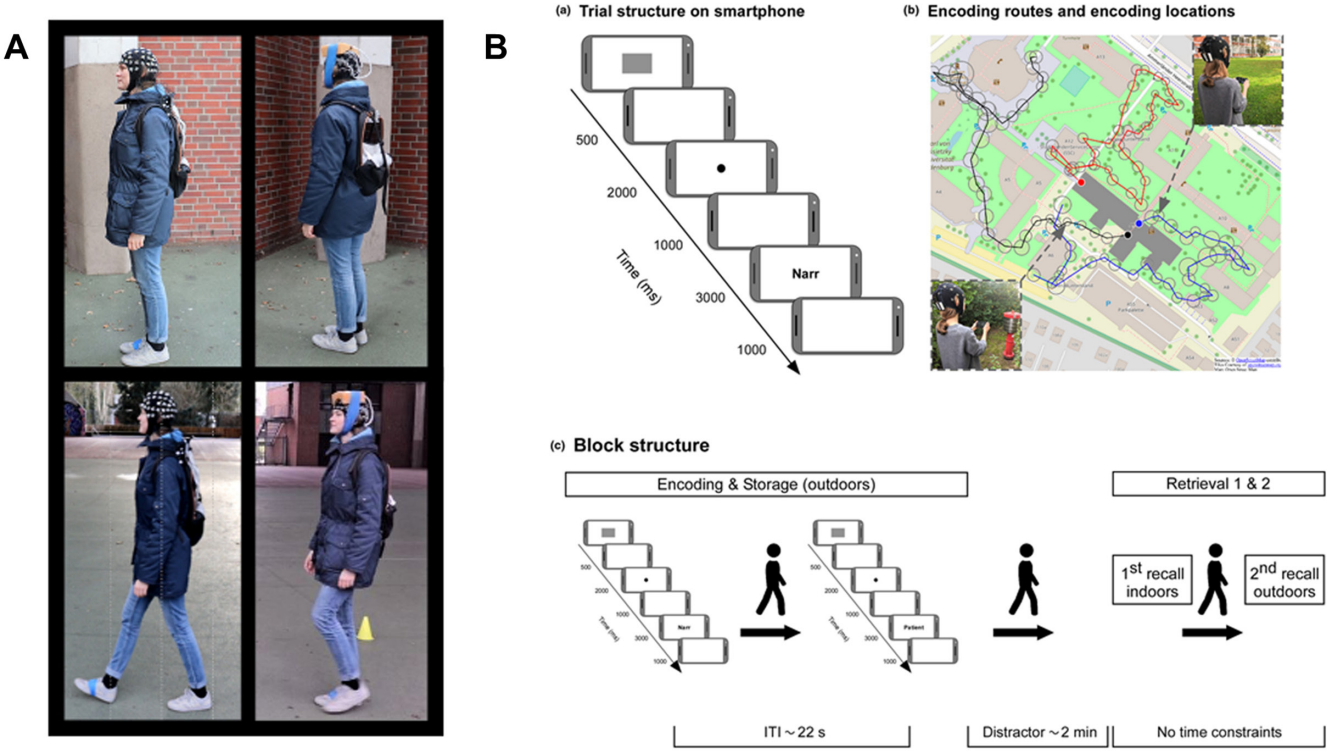


FIGURE 4 (A) Walking EEG paradigms. Scanlon et al. (2020). (B) Piñeyro Salvidegoitia et al. (2019).

In order to ameliorate the mechanical artifacts from the skin sensor interface, sensors made out of wearable and flexible materials have been previously proposed as an alternative to rigid electronics. For example, along with collaborators in material engineering, we have proposed a thin breathable/stretchable sensor design that can be attached to the skin and reused repeatedly (Jang et al., 2014). These flexible electrodes were used to measure EEG in a driving simulation paradigm to demonstrate the feasibility of these systems in real tasks. Some flexible electrodes have even been integrated with full flexible circuit boards to create mobile electronic solutions (Xu et al., 2014). Likewise, we have proposed a two-layered skin-adhering electrode for wearable healthcare

14698986, 2024, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/psyp.14603, Wiley Online Library on [24/08/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

and research purposes (La et al., 2018). We showed that these two-layered e-textile electrodes showed fewer motion artifacts relative to a commercial sensor. More recently, other sensors such as the Flex-printed forehead EEG sensors (fEEGrid) have been proposed (Blum et al., 2020). This 24-channel system can be attached to the forehead and face areas for long term-recording. The authors observed that participants did not report discomfort while using this system.

Even with state-of-the-art electrodes, EEG recordings are greatly vulnerable to motion artifacts (Jiang et al., 2019). This means that even when researchers take innovative approaches to record data during motion, there is a constant need to rely on data cleaning techniques to improve the quality of the results, the topic of the next section.

2.5 | Artifact correction techniques

To account for the influence of artifacts in the EEG data researchers have favored several available methods aimed at identifying and correcting noise artifacts. Artifact correction techniques not only enhance the signal clarity of mobile EEG data but also bolster the validity of measures obtained from the data. The effectiveness of Independent Component Analysis (ICA) in isolating genuine neural signals from artifacts underpins the construct validity of EEG measures in naturalistic settings. Similarly, techniques like Artifact Subspace Reconstruction (ASR) contribute to face validity by ensuring that the EEG data recorded in dynamic environments align with the known characteristics of brain activity established in controlled laboratory conditions.

Perhaps one of the most encountered artifact correction techniques in MoBI papers is the independent component analysis (ICA; Bell & Sejnowski, 1995). ICA is applied to divide data into different components based on statistical independence (Luck & Kappenman, 2016). In the context of EEG data, ICA is highly effective at separating eyeblinks, mechanical artifacts, and other sources of noise from the EEG signal (Wunderlich & Gramann, 2020). From the early stages of MoBI research, it has been argued that ICA is an effective algorithm for the separation of movement artifacts from the continuous EEG data during motion (Makeig et al., 2009). Moreover, this method can spatially separate the sources of data (mechanical artifacts, eye movements, etc.) to locate the signal projections recorded at the scalp (Gramann et al., 2011). The ability to separate these signals during active behavior is one exciting aspect of data analysis afforded by the MoBI approach.

ICA has been successfully applied to remove a wide range of artifacts from the EEG signal during naturalistic behavior. Some of these include eye and neck artifacts during slow and fast treadmill walking (Gramann et al., 2010). However, Gramann found that although ICA is effective for walking-induced artifacts, large artifacts induced by running could not be removed using ICA. Furthermore, ICA has been used to remove blink artifacts during outdoor walking (Debener et al., 2012; Gabard-Durnam et al., 2019; Jacobsen et al., 2021; Scanlon et al., 2022), mechanical and cable artifacts during walking (Malcolm et al., 2018), and mechanical artifacts during obstacle courses (Reiser et al., 2019). In a simulated work environment where participants moved boxes between workstations ICA was used to remove body artifacts during this physical task (Wascher et al., 2016). This method has also been applied to separate biological signals that affect EEG such as heartbeat artifacts (Ladouce et al., 2019). ICA has also been employed to remove blink artifacts during cycling (Storzer et al., 2016; Zink et al., 2016). Further, Zink et al. (2016) have proposed using Canonical Correlation Analysis (CCA) to remove muscle artifacts. This technique uses autocorrelations to separate EEG data (high autocorrelation) from artifacts (low autocorrelation). These authors used a CCA in a cycling paradigm and argued that this method separates artifacts better than the ICA.

Other mobile studies have also adopted different methods for artifact removal. For example, Dehais et al. (2019) used Artifact Subspace Reconstruction (ASR) to remove motion artifacts during real flying due to the reduced number of electrodes used. The ASR algorithm is based off of a Principle Component Analysis (PCA), and is first calibrated on clean data to create statistical cutoffs for observed component variance during the to-be-corrected-data (Mullen et al., 2013). ASR correction has also been used for motion artifacts during treadmill walking at different speeds (Nathan & Contreras-Vidal, 2016). The authors suggested this method is most effective at dealing with muscle bursts and movement.

Finally, work by Scanlon et al. (2017, 2020; Scanlon, Townsend, et al., 2019) suggests that even simple methods such as the regression-based eye movement correction procedure (Gratton et al., 1983) can be suitable for some mobile paradigms. It was demonstrated that with enough trials, this method can yield high-quality data during stationary cycling (Scanlon et al., 2017), and even in outdoor environments such as parks (Scanlon et al., 2020) and local bike lanes (Robles et al., 2022). It is important to consider that relative to walking and running bicycle riding is a smoother task where the signal might not be as contaminated by large artifacts. Generally, ICA is known to be more robust than regression-based methods

14698986, 2024, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/psyp.14603, Wiley Online Library on [24/08/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

under less ideal data collection situations (Hoffmann & Falkenstein, 2008), however, these cycling studies show that good quality data can be achieved during motion without using more computationally heavy decomposition methods. As we continue to push the boundaries of mobile EEG research, it is essential to address the remaining challenges and explore new advancements and future directions.

2.6 | Advancements and future directions in mobile EEG research

Given the challenges that exist in recording EEG in motion (Ladouce et al., 2017), it is clear that it will take many years to achieve full mobility during data collection (e.g., running, high-performance sports, etc.). For example, our workgroup has focused on the creation of suitable behavioral paradigms for motion experiments. This includes the validation of portable computers for mobile paradigms (Kuziek et al., 2017), designing a task for navigation in unstructured environments, using a backpack to facilitate natural motion during EEG collection, and designing the hardware for button pressing (Robles et al., 2021; Scanlon et al., 2020). Furthermore, to this day, mobile and naturalistic paradigms are characterized by a higher degree of data loss relative to stationary laboratory studies (Ladouce et al., 2019). For this reason, advancements in data collection sensors and methods will need to take place to advance mobile neuroscience beyond its current limitations. Although this is certainly a major obstacle in the study of naturalistic cognition, many methodological recommendations can be found in the literature to achieve good signal quality.

From early MoBI publications, several guidelines have been proposed for developing future mobile paradigms. These include using brain and muscle sensors to record behaviors in 3D environments, using lightweight or wireless sensors with battery-powered receivers, and recording subject interactions with the environment (Gramann et al., 2014; Jungnickel et al., 2018; Makeig et al., 2009). Other recommendations include analyzing EEG and behavioral data as well as using decomposition techniques to account for artifacts (Gramann et al., 2011). In a study comparing different EEG systems during walking, Oliveira et al. (2016) suggested using a 75 μV rejection threshold as well as an analysis of pre-stimulus noise, signal to noise ratio, and P3 variance. The authors noted that these are important metrics to assess overall data quality during motion. They proposed using an average of 70 trials for optimal P3 recording and found that wet systems yield higher quality and are better tolerated by subjects. Likewise, Scanlon et al. (2017) conducted a power analysis to estimate the number of trials needed to attain 80% replications of the observed mean P3 difference between cycling and sitting still. Mobile recording required increased number of trials to produce statistical power similar to indoor conditions. For example, to accurately record the MMN in the bike, mobile data required 25 targets and 140 trials relative to the 20 targets and 80 standard trials in the indoor cycling condition.

Using Web of Science, a global search for the term "mobile brain-body imaging" showed that since the term was first coined in 2009, the number of publications in this field has been growing steadily through the years (except for 2020 where research worldwide has been affected). Although most of the articles reviewed in the present work highlight current challenges in the field, it is clear that mobile research is growing at a steady pace. One important consideration is that to our knowledge there are no systematic reviews in the field. Many research parameters can be illustrated in systematic studies that can help mobile EEG researchers plan future paradigms (Lau-Zhu et al., 2019). As with any emerging field, there are contrasting views regarding the best methodology to collect EEG in motion. However, with the current growth in the mobile EEG literature, and with access to increasingly cheaper and better portable technology, we can predict an expansion in EEG paradigms beyond the laboratory over the next few years.

2.7 | Section summary

In summary, this section has laid the groundwork for understanding mobile EEG research, underscoring the importance of electrode selection and artifact correction in enhancing the validity of findings. We have examined how these methodological choices affect construct and face validity, ensuring that our insights into attention and sensory processing in natural settings are both reliable and generalizable. The subsequent section will delve into the practical applications of these validated tools, showcasing their contributions to novel discoveries in cognitive function. Building on the methodological foundations established in this section, we will now explore how mobile EEG research is shedding light on cognitive processes in real-world settings, focusing on studies on selective attention.

3 | EXPLORING SELECTIVE ATTENTION IN REAL-WORLD SETTINGS WITH MOBILE EEG

With developments in portable methodology researchers have been interested in studying cognitive phenomena

14698986, 2024, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/psyp.14603, Wiley Online Library on [24/08/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

while subjects are in motion. Understanding how attention operates in naturalistic scenarios is of paramount importance as it can shed light on real-world cognitive processes. In the current section, we are discussing several studies where subjects were engaged in naturalistic tasks such as walking and cycling to highlight some contributions of mobile EEG research to the field of attention. One key aspect of attention that has been investigated using mobile EEG is the neural mechanisms underlying attentional processes, one measure of which is the P3 component of the ERP.

3.1 | Examining the P3 component in mobile EEG studies of human attention

EEG can be used to record several indices of attention such as the P3. The amplitude of this ERP component has been previously linked to cognitive resources devoted to experimental tasks (Polich & Comerchero, 2003). Generally, a reduction in P3 voltage is associated with increased task efforts which could reflect the reallocation of limited cognitive resources (Kok, 2001). Since this particular component has wide replicability in traditional laboratory settings (Zamrini et al., 1991), mobile studies have explored the P3 under less structured research paradigms, and while participants perform physical tasks. One central assumption in these studies is that there are shared resources between areas of motor control and cognitive processing (Leone et al., 2017). For instance, a complex motor task might draw cognitive resources in ways similar to a traditional challenging computer task. By measuring changes in P3 amplitude it could be possible to measure a relative index of cognitive effort while in various types of motion.

De Vos et al. (2014) used an oddball task to record the P3 from subjects during sitting and walking respectively. They found a 30% reduction of the P3 amplitude in the walking condition that could not be accounted for by differences in noise levels between conditions. The authors argued that the overall level of sensory/cognitive engagement during walking could be creating these differences in attentional allocation. In a study using a Go No/Go task during walking and standing, researchers found a lower P3 amplitude during the walking condition reflecting differences in cognitive processing while participants were in motion (De Sanctis et al., 2014). The authors concluded that the observed differences in the EEG signal represent an adaptive mechanism to improve performance during walking (hence the lower P3 amplitude during motion). Similar results were obtained by our group in a walking study where subjects completed an oddball task walking and standing outdoors (Scanlon et al., 2021). We found a reduction in P3 amplitude during the walking condition, concluding that the reduction in P3 amplitude could be due to sensory processing and motor activity in the walking condition.

In a study manipulating motor difficulty while subjects completed a cognitive task authors found that the both cognitive and motor processes independently reduced P3 amplitude (Liebherr et al., 2018). These results are supported by a recent study where participants completed an oddball task during standing, walking, and completing an obstacle course (Reiser et al., 2019). In this study, researchers observed decrease in the P3 amplitude during the moving conditions which authors interpreted as a reduction in cognitive resources during the more complicated motor tasks. Additionally, they found an event-related decrease in theta power during motion that suggests a decrease in sustained attention during the motion conditions. In an oddball study where subjects either stood or walked while completing an oddball task, authors found that the reduction in P3 amplitude was driven by both inertial and visual processing more so than by increases in motor demands (Ladouce et al., 2019). Taken together, these studies show the feasibility of using ERP components to explore differences in information processing in motion. In particular, these studies demonstrate the utility of using the P3 as an index of task effort beyond the laboratory. This provides new insights into resource allocation under more dynamic paradigms.

Mobile paradigms have also been used to explore differences in cognitive processing between young and old adults during walking. For example, using a visual task where participants respond to peripheral flashing lights, authors found differences in early visual processing between young and older subjects as measured by P1 latency and amplitude (Protzak et al., 2021). Interestingly, Protzak et al. (2021) found that only young participants exhibited a reduced P3 amplitude during a walking task compared to sitting, whereas both age groups showed longer latency with more motor task complexity. Interestingly however, this study failed to show any covariation between visual processing performance and P3 features amplitude or latency. Ultimately, as the body of mobile EEG literature grows, increased validity will clarify some of the disparate results and separate null findings from inability to accurately measure signal amongst noise, refining and redefining our understanding of ERP components. To further advance our understanding of attentional mechanisms in real-world settings, researchers have begun exploring innovative approaches that move beyond traditional EEG markers such as physiologically locked averaging, and oscillatory investigations.

14698986, 2024, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/psyp.14603, Wiley Online Library on [24/08/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

3.2 | Innovative approaches to investigating cognitive processes in outdoor environments

In looking for markers of attention while outdoors, researchers have proposed an alternative approach using markers such as eyeblinks and saccades to investigate attentional mechanisms during motion. Using a task-free paradigm, investigators manipulated walking speed as well as lighting levels and found that reductions of alpha power recorded over the occipital region were independent of visual stimulation during walking (Cao et al., 2020). These findings are in line with previous studies showing an increase in cortical excitability during walking, though these authors argued that the modulation in alpha power is not driven by visual sensory input during walking. It was also found that walking increases the rate of saccades and eyeblinks during the supporting phase of the gait cycle. Interestingly, this increase in eye movements was found to occur during the walking phase where visual processing is decreased. The authors concluded that this walking phase-dependent modulation of eye movement could serve as a preparatory mechanism for the gait phase where visual processing is most optimal. This approach could open new avenues to explore attentional mechanisms during unconstrained behavior such as walking outdoors.

One of the most crucial challenges in doing outdoor experiments is the large overlap of ERP components that is produced by the bombardment of the senses. In an attempt to overcome some of these issues, Wunderlich and Gramann (2020) have also proposed using meaningful eye and saccade-related brain potentials to extract events from outdoor situations. They demonstrated that these eye movements are tied to attentional processes such as information allocation. Using these blink-saccade-related potentials, the authors found that subjects displayed higher levels of cognitive engagement during landmark-based instructions relative to standard navigation instructions. By using this less traditional approach authors showed that brain dynamics can be captured in a city environment. This represents an advancement in EEG data collection given that random events that occur outdoors cannot be easily be identified during data analysis using traditional event triggers.

Other mobile studies have used alternative approaches to study mental processes. For example, using outdoor landmarks, (Piñeyro Salvidegoitia et al., 2019) deployed a memory task on a university campus (Figure 4B). They found increased memory recall associated with landmark features while outdoors. They also found differences in ERP components and EEG frequency spectra associated with better memory encoding. In a MoBI study, (Pizzamiglio et al., 2017) compared participants' gait performance while self-paced walking, walking while chatting, and walking while texting. They found modulations in the theta and beta frequencies related to walking and conversing, while walking and texting were marked by changes in beta power. These changes were accompanied by a decrease in gait performance in dual-tasking. We have also a hand-controller-operated electric skateboard where subjects completed an oddball riding on a smooth, running-track surface to minimize data noise (Robles et al., 2021; Figure 5). The authors demonstrated that this novel paradigm was feasible for recording reliable P3 and EEG frequency spectra during free navigation while minimizing body movements associated with walking. These findings demonstrate the feasibility of mobile EEG in capturing attentional mechanisms during motion, which is further explored in the context of cycling paradigms in outdoor settings. Cycling, in particular, offers a unique opportunity to investigate attentional processes in a real-world setting, as it requires the simultaneous coordination of physical and cognitive resources, making it an ideal paradigm for understanding attentional mechanisms in motion.

3.3 | Understanding attention mechanisms through cycling paradigms in outdoor settings

With increases in hardware portability researchers have employed cycling paradigms to explore attentional mechanisms in motion. For example, in a study using a three-class auditory oddball task, researchers were able to quantify the oddball P3 while participants were either sitting still on a stationary bicycle, pedaling the stationary bicycle on a stand, or moving freely on the bike through the world (Zink et al., 2016). Consistent with many walking studies, researchers found a decreased P3 amplitude and a decrease of alpha power during active cycling relative to the stationary conditions. Crucially, this difference in resource allocation remained after researchers controlled for muscle artifacts. Both P3 and alpha reductions were interpreted as the result of an increase in cognitive demands and task difficulty. Interestingly, in a study comparing cortical power modulations between bicycling and walking, Storzer et al. (2016) found that walking is associated with a larger decrease in alpha power. Decreases in alpha are generally linked to increases cortical excitability (Klimesch et al., 2007), and here was attributed to higher motor planning and sensory processing during walking. Storzer concluded that relative to cycling, walking requires higher cortical activation, and this difference might be driven by the monotonous nature of cycling.

14698986, 2024, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/psyp.14603, Wiley Online Library on [24/08/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

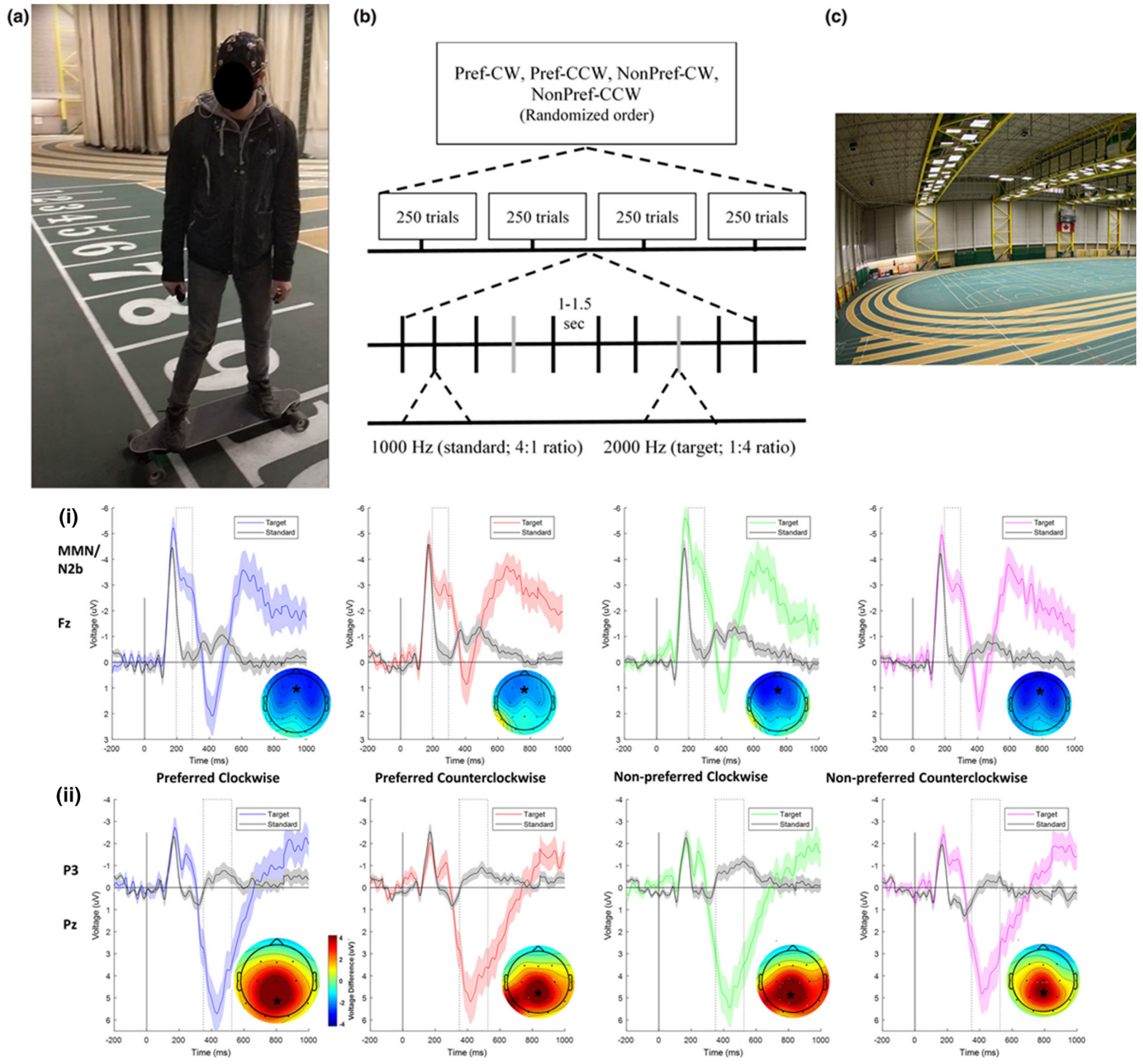


FIGURE 5 Skateboard EEG Paradigm. Robles et al. (2021). (A) Participant with backpack and EEG system riding electric skateboard in the University Pavilion (Butterdome) at the University of Alberta. (B) Experiment structure for auditory oddball tasks during skating. (C) View of the 200 meter indoor track. Bottom: (i) Tone elicited ERPs at Fz for target and standard tones for the four skateboard stance conditions: Clockwise, counterclockwise, with both the preferred and non-preferred skateboard stance. Topographic maps of the difference between target and standard in the indicated time window. (ii) Tone elicited ERPs at Pz for target and standard tones for the four skateboard stance conditions with topographies of the difference.

In a series of cycling studies Scanlon and colleagues have shown that relative to indoor sitting, outdoor cycling is associated with a more negative P3 and a reduction of alpha power (Scanlon, Townsend, et al., 2019). The authors suggested that the decrease in P3 amplitude outdoors reflects the reallocation of cognitive resources between the cognitive task and the busier, outdoor cycling. Additionally, the overall decrease in alpha power during outdoor cycling was attributed to the sensory overload brought about by the busy outdoor environment. Interestingly, the authors found an unexpected increase in the N1 component during the outdoor cycling condition. This increase in voltage was interpreted as a boost in auditory filtering required to maintain good performance in the auditory oddball while outdoors. This effect was replicated in the lab using naturalistic background sounds simulating the busy street (Scanlon, Cormier, et al., 2019) In a follow-up study, we deployed the cycling paradigm to different outdoor

14698986, 2024, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/psyp.14603, Wiley Online Library on [24/08/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

environments differing in auditory noise levels (Scanlon et al., 2020). Consistent with previous work, it was found that the N1 was increased when participants rode the bicycle in the busier cycling environment. It was concluded that this N1 modulation by environmental noise represents an enhancement in sensory processes during loud environments to maintain task performance. In conclusion, the cycling paradigm has demonstrated the ability of mobile EEG to measure behavioral and neural attentional effects during real-world attention in complex tasks.

3.4 | Section summary: Insights from mobile EEG research on selective attention in real-world environments

The reviewed studies reveal several points regarding the contributions of mobile EEG to the field of selective attention. First, they show that many of the factors that make outdoor research challenging (various sources of artifacts, low environmental control, etc.) do not necessarily compromise the validity and reliability of studies if suitable paradigms are developed. Even when there are no gold standard benchmarks in mobile research, these studies demonstrate that it is possible to transition studies beyond the stationary laboratory into less structured environments without sacrificing validity and reliability. Furthermore, we need to consider the question of whether outdoor studies validate or contradict the large body of knowledge accrued over decades of indoor research. By showing specific task modulations of attentional components originally validated in laboratory-based studies (e.g., alpha power, P3, etc.), mobile paradigms serve as evidence that the body of knowledge gained over decades is valid in highly ecologically valid settings. An important point raised by Reiser et al. (2019) and others is that high ecological validity should not compromise the quality of the signals collected during mobile studies. This ability to transition laboratory research into portable/outdoors setting should provide experienced researchers an opportunity to validate paradigms under novel settings.

Considering the biological and functional complexity of the brain and how that imposes a challenge for cognitive neuroscientists (Poldrack et al., 2011), the mobile EEG literature reviewed here represents an important step in understanding attentional processes under unstructured real-world environments. Many mobile studies so far complement laboratory research even though the traditional laboratory approach is very restrictive creating a limited subject experience (Ladouce et al., 2017). This opens new and exciting avenues for research where laboratory-based studies can serve as optimal control to then explore cognitive processes under less structured environments. If there were no replicability or proper levels of validity and reliability during mobile studies, researchers would have brought into question the degree to which laboratory studies were portraying an accurate picture of brain processes.

In summary, the exploration of selective attention through mobile EEG is a burgeoning field that has revealed promising insights. Although mobile EEG is a relatively new area of research, the reviewed studies have showcased its potential by successfully replicating classic laboratory findings in the context of realistic scenarios. These investigations have not only validated our existing knowledge but have also illuminated the challenges and cognitive costs associated with engaging in complex tasks within naturalistic settings, all quantifiable through traditional EEG techniques. The implementation of mobile paradigms has become essential for unraveling the intricacies of human information processing in motion. As we delve into the next section, we will explore the exciting future directions and advancements in this field, where the development of portable research tools promises fresh paradigms across a broad range of applications and fields such as rehabilitation medicine and education.

4 | EXPANDING HORIZONS: MOBILE EEG APPLICATIONS ACROSS DIVERSE FIELDS

One of the greatest benefits of the development of portable EEG technology is that it can greatly benefit society in several areas such as rehabilitation medicine, workplace/driving safety. For example, by understanding the cortical dynamics associated with walking researchers can develop future rehabilitation approaches (Seeber et al., 2015; Wagner et al., 2012). This can be accomplished by analyzing different frequency bands signal (theta, alpha, etc.) that make up the raw EEG. There are several functional correlates associated with the activity of these frequencies such as information processing, and attentional states (Lopes da Silva, 2013). For example, changes in the theta and beta band have been associated with increases in mental fatigue (Craig et al., 2012). When we consider mental states such as fatigue, reliable quantification of these mental states via portable EEG could be very promising. This potential for quantifying mental fatigue and cognitive workload is explored next.

4.1 | Assessing mental fatigue and cognitive workload with mobile EEG

Previous mobile research has shown that mental fatigue can be quantified via EEG. For example, using a cognitive

14698986, 2024, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/psyp.14603, Wiley Online Library on [24/08/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

task, Wascher et al. (2014) employed a four-hour paradigm to measure changes in EEG frequencies over time. Relative to the start of the experiment, they found an increase in frontal theta and occipital alpha towards the last blocks. The authors argued that the observed changes in the theta band could be related to a depletion of executive control and higher error rates. Likewise, Wascher et al. (2016) measured the cognitive and subjective effects of workload fatigue between old and young adults while they performed various cognitive tasks plus a physical task (moving boxes). The impact of task monotony in young participants was marked by an increase in alpha power and error rates. The authors attributed this increase in alpha activity to attentional withdrawal. An important aspect of the methodology employed in this study is that they used eyeblink-related analyses to measure event-related synchronizations (ERS) in the frequency bands of interest. This shows that it is possible to capture changes in cognitive demands and fatigue using blink-related activity.

Mobile EEG has made it possible to study cognitive processes under ecologically valid conditions such as driving and piloting. This can be beneficial when it comes to identifying and preventing road and workplace-related accidents generally attributed to increases in cognitive load and fatigue (Zhao et al., 2012). For instance, simulation driving experiments have found correlations between alpha and theta frequencies and the incidence of driving errors that include lane deviations and crashes (Borghini et al., 2014). More recently, Di Flumeri et al. (2018) recorded subjects' EEG while they drove under real-world conditions (driving under rush hour vs normal hours). They used road complexity and traffic volume as relative indicators of mental workload. They compared the ratio of frontal theta and parietal alpha during these two conditions and found an increase of both band powers in the rush hour relative to the normal hours. They argued that the EEG signal is more sensitive in measuring mental workload relative to eye tracking and subjective measures. They also staged several events during driving and used them as data markers (i.e., pedestrian crossing the road and a car entering the lane).

In another feasibility study, Dehais et al. (2019) used an auditory oddball task to test for cognitive load in pilots flying under high and low workload conditions. The authors found a reduced P3 amplitude and changes in alpha and theta power during the high workload task (when pilots operated the airplane). The authors concluded that the dry electrode portable EEG system could be used to assess mental workload under real flying conditions. A second goal of the study was to use single-trial classification of the EEG signals to distinguish between the low and high workload conditions. They found that the classification accuracy was not significant using the P3 signal but found a much higher classification accuracy using the EEG frequency spectra. The findings on mental fatigue and cognitive workload have important implications for the development of Brain–Computer Interfaces (BCIs) for driver monitoring and rehabilitation, which is discussed in the next section.

4.2 | Empowering brain–computer interfaces (BCIs) for driver monitoring and rehabilitation

In addition to identifying some neural markers of cognitive load, there are very promising applications of mobile EEG such as Brain–Computer Interfaces (BCI). BCI works by creating a model of neural activity under a specific task or cognitive state and this signal classification can then be applied to a variety of outputs (Ojeda et al., 2014). For instance, BCI allows for the identification of functional brain signals to achieve voluntary control over machines without physical contact (Vallabhaneni et al., 2005). BCI methods have used alpha power changes to control devices, but also during car driving and simulated piloting (Borghini et al., 2014).

When it comes to driving and accident prevention, BCI systems can be used to monitor errors while driving. This error monitoring has potential applications for emergency braking systems which could even activate before the subject physically touches the brake pedal (Blum et al., 2017). Additionally, BCI models have been used to identify increases in alpha power resulting from the transition from eyes open to eyes closed (Lim et al., 2014). By identifying a closed-eyes alpha state the authors proposed a drowsiness detection system that prevents drivers from falling asleep. It does so by detecting increases in alpha power induced when people maintain their eyes closed for certain periods. This drowsiness detection model was set to trigger an alarm in the event a driver experiences an increase in low alpha activity associated with the transition between eyes open and eyes closed (Lim et al., 2014).

These applications of EEG and BCI systems have obvious implications for road safety and accident prevention. Additionally, this technology can aid patients with physical disabilities (e.g., controlling prosthetic limbs) by training models during the rehabilitation phase (Gramann et al., 2011). Since the MoBI approach focuses on capturing brain dynamics under naturalistic conditions, the development of naturalistic paradigms will be greatly beneficial for rehabilitation medicine that includes gait, and other motion-related behaviors (Gramann et al., 2014). It could be a matter of time until portable paradigms achieve an ideal signal quality that leads to higher classification

14698986, 2024, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/psyp.14603, Wiley Online Library on [24/08/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

accuracy for prosthetic control. The potential of BCIs for driver monitoring and rehabilitation is just one example of the many applications of mobile EEG technology, which are explored in the next section.

4.3 | Expanding horizons: Diverse applications of mobile EEG technology

There are several other areas of research that can benefit from the advancements in mobile EEG technology. In a review article, Lau-Zhu et al. (2019) proposed the use of mobile EEG for neurodevelopmental studies. The authors argued that mobile technology can now be used to study a range of cognitive processes related to neurodevelopment that include memory and information processing. They also suggested that due to the increased flexibility afforded by portable EEG, this technology is more suitable for children (e.g., miniature and concealed electrode forms), and also for large-scale studies and novel paradigms. With increasing EEG data collection in remote parts of the world, researchers can gain valuable information regarding EEG markers of child development (Alahmadi et al., 2016; Giertuga et al., 2017; Leal Neto et al., 2021). Likewise, along with colleagues in medicine, we used a Muse headband to record data from ischemic stroke patients (Wilkinson et al., 2020). We were able to identify moderate to severe stroke patterns in the EEG data using machine learning. Mobile paradigms can also be deployed to real-world situations such as school settings. For instance, Dikker et al. (2017) used a mobile paradigm to record continuous data from a classroom of students over an entire semester period to explore brain synchrony during classroom interactions.

Advancements in Bluetooth technology now allow EEG signals to be collected over webpages such as http://www.eegedu.com/. Tutorials showing how to collect EEG using Bluetooth connections with devices are also now available (Castillo, 2016). Additionally, android-based collection (Blum et al., 2017; Debener et al., 2015) greatly increases the accessibility of research by employing consumer-grade smartphones. Other scientific open source initiatives such as NeuroTechEdu (http://learn.neurotechedu.com/) have been built to form a scientific community to collaborate in mobile EEG, BCI, and other research techniques. Resources such as MNE (https://mne.tools/stable/index.html) offer free imaging packages where students and enthusiasts can learn to analyze and visualize EEG and other signals using Python. Taken together, these advancements in low-cost-portable technology, EEG hardware, and open resources can greatly benefit university students and researchers who can learn to collect and analyze EEG data without having to acquire expensive hardware.

4.4 | Section summary

In summary, the development of portable EEG technology has led to substantial breakthroughs in understanding attention and cognitive processes. Its applications span various domains, including quantifying mental fatigue and cognitive workload, enhancing Brain–Computer Interfaces (BCIs) for driver monitoring and rehabilitation, and expanding its use in neurodevelopmental studies and real-world settings. This technology has far-reaching implications for improving cognitive performance, safety, and overall well-being.

Mobile EEG technology offers real-time insights into cognitive states, making it invaluable for workplace and driving safety. BCIs enabled by mobile EEG hold potential for machine control and aiding individuals with physical disabilities. Additionally, it facilitates research in child development and adapts to real-world scenarios, democratizing EEG research through open-source initiatives and accessible technology. As mobile EEG technology continues to advance, its impact on attention research and broader applications is poised to grow exponentially, leading to innovative solutions and improved outcomes.

By leveraging mobile EEG technology, researchers can now study cognitive processes in naturalistic environments, overcoming the limitations of traditional laboratory-based research. This enables a more comprehensive understanding of attention and cognitive processes in real-world settings, ultimately leading to more effective interventions and applications. For instance, mobile EEG has already shed light on the neural mechanisms underlying attention and cognitive fatigue, informing strategies to mitigate their effects.

5 | OVERALL CONCLUSION

The present review explored the developing field of portable EEG. We discussed the limitations in laboratory research leading to the need to use mobile paradigms to study cognitive and mental processes away from the laboratory, as well as the developments that allowed increasingly naturalistic mobile EEG recordings to be made. We introduced several new alternatives in consumer-based hardware and discussed several studies that highlight their benefits as well as some of their current limitations. We then discussed the feasibility and methodological challenges that are still present in the mobile literature. We conclude that these advancements continue to improve the validity and reliability of mobile EEG research and provide valuable new insights in their novel applications across fields, ultimately strengthening the field of cognitive neuroscience.

14698986, 2024, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/psyp.14603, Wiley Online Library on [24/08/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Furthermore, we looked at the contributions of mobile EEG in the field of attention by reviewing studies that explored resource allocation while subjects completed active tasks such as walking and cycling. The reviewed studies demonstrate that mobile research can effectively replicate laboratory-based research, and this is an important contribution to the field of cognitive neuroscience. This is due to the ability to further replicate long-used paradigms under less structured environments. Additionally, it was argued that instead of invalidating traditional EEG research, mobile studies complement laboratory-based research. The ability to transition cognitive studies away from the laboratory opens exciting avenues for the investigation of cognitive phenomena using new paradigms.

In this review, we have seen how advancements in mobile EEG technology are not just expanding the boundaries of where and how we can conduct neuroscience research, but are also strengthening the construct, predictive, and face validity of our findings. By bringing cognitive studies into naturalistic environments, we align our experimental paradigms more closely with the real-world phenomena they aim to explain, enhancing the ecological validity of our work. As a result, we can increase the generalizability and applicability of our findings to real-world situations. As we continue to refine mobile EEG methods and tackle the remaining methodological challenges, we enhance our capacity to capture accurate representations of brain activity in dynamic settings. This progress is pivotal in ensuring that our research outcomes are not only innovative but also robust and applicable in a variety of practical, everyday contexts.

We finished the review by highlighting several promising applications of mobile EEG in areas such as road safety and rehabilitation medicine. Portable research tools offer new opportunities to quantify brain signals associated with increased task load and fatigue. Additionally, BCI methods can be used with portable EEG settings to quantify functional signals that can facilitate physical control in patients with limited mobility. Notably, mobile EEG technology enables researchers to study cognitive processes in naturalistic environments, providing insights into real-world brain function and behavior. Also, advancements in Bluetooth data collection and open-source analysis packages can bring EEG to the classroom where students can collect and analyze data using relatively low-cost gear. Overall, the future of mobile EEG is very promising, and its potential to improve our understanding of brain function and behavior in real-world settings is vast. Although many of the applications discussed in this review are still under development, the future of mobile EEG is very promising. Moreover, mobile EEG technology has the potential to revolutionize cognitive neuroscience research, enabling us to study brain function and behavior in unprecedented ways. Its impact could be transformative, leading to breakthroughs in healthcare, education, and transportation, and informing more effective interventions, treatments, and policies. As mobile EEG technology continues to evolve, we envision a future where it becomes an indispensable tool for understanding human cognition and behavior, ultimately improving human lives and society as a whole.

AUTHOR CONTRIBUTIONS

Kyle E. Mathewson: Conceptualization; data curation; funding acquisition; investigation; methodology; project administration; resources; software; supervision; validation; writing – review and editing. **Jonathan P. Kuziek:** Conceptualization; data curation; formal analysis; investigation; methodology; resources; software; visualization; writing – review and editing. **Joanna E. M. Scanlon:** Conceptualization; data curation; formal analysis; investigation; methodology; resources; visualization; writing – original draft. **Daniel Robles:** Conceptualization; data curation; formal analysis; investigation; methodology; project administration; validation; visualization; writing – original draft; writing – review and editing.

CONFLICT OF INTEREST STATEMENT

None.

DATA AVAILABILITY STATEMENT

All data and materials reported in this manuscript from our lab is available either online currently or upon request. General tools used by our lab can be found at https://github.com/APPLabUofA and at https://github.com/kylemath.

ORCID

Kyle E. Mathewson https://orcid.org/0000-0003-4856-3980
Jonathan P. Kuziek https://orcid.org/0000-0003-1368-9102
Daniel Robles https://orcid.org/0000-0002-0065-6224

REFERENCES

Alahmadi, N., Evdokimov, S. A., Kropotov, Y. J., Müller, A. M., & Jäncke, L. (2016). Different resting state EEG features in children from Switzerland and Saudi Arabia. *Frontiers in Human Neuroscience*, *10*, 559. https://doi.org/10.3389/fnhum.2016.00559

Al-Yahya, E., Dawes, H., Smith, L., Dennis, A., Howells, K., & Cockburn, J. (2011). Cognitive motor interference while walking: A systematic review and meta-analysis. *Neuroscience and Biobehavioral Reviews*, *35*(3), 715–728. https://doi.org/10.1016/j.neubiorev.2010.08.008

Bell, A. J., & Sejnowski, T. J. (1995). An information-maximization approach to blind separation and blind deconvolution. *Neural*

14698986, 2024, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/psyp.14603, Wiley Online Library on [24/08/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Computation, *7*(6), 1129–1159. https://doi.org/10.1162/neco.1995.7.6.1129

Bleichner, M. G., & Debener, S. (2017). Concealed, unobtrusive ear-centered EEG acquisition: Ceegrids for transparent EEG. *Frontiers in Human Neuroscience*, *11*, 163. https://doi.org/10.3389/fnhum.2017.00163

Blum, S., Debener, S., Emkes, R., Volkening, N., Fudickar, S., & Bleichner, M. G. (2017). EEG recording and online signal processing on android: A multiapp framework for brain-computer interfaces on smartphone. *BioMed Research International*, *2017*, 1–12. https://doi.org/10.1155/2017/3072870

Blum, S., Emkes, R., Minow, F., Anlauff, J., Finke, A., & Debener, S. (2020). Flex-printed forehead EEG sensors (fEEGrid) for long-term EEG acquisition. *Journal of Neural Engineering*, *17*(3), 034003. https://doi.org/10.1088/1741-2552/ab914c

Borghini, G., Astolfi, L., Vecchiato, G., Mattia, D., & Babiloni, F. (2014). Measuring neurophysiological signals in aircraft pilots and car drivers for the assessment of mental workload, fatigue and drowsiness. *Neuroscience and Biobehavioral Reviews.*, *44*, 58–75. https://doi.org/10.1016/j.neubiorev.2012.10.003

Cao, L., Chen, X., & Haendel, B. F. (2020). Overground walking decreases alpha activity and entrains eye movements in humans. *Frontiers in Human Neuroscience*, *14*, 561755.

Castillo, A. (2016). Muse 2016 headband + web bluetooth. Medium. https://medium.com/@castillo.io/muse-2016-headband-web-bluetooth-11ddcfa74c83

Çınar, S. (2021). Design of an automatic hybrid system for removal of eye-blink artifacts from EEG recordings. *Biomedical Signal Processing and Control*, *67*, 102543. https://doi.org/10.1016/j.bspc.2021.102543

Clark, A., & Chalmers, D. (1998). The extended mind. *Analysis*, *58*(1), 7–19. https://doi.org/10.1093/ANALYS/58.1.7

Cohen, M. X. (2017). Where does EEG come from and what does it mean? *Trends in Neurosciences*, *40*(4), 208–218. https://doi.org/10.1016/j.tins.2017.02.004

Craig, A., Tran, Y., Wijesuriya, N., & Nguyen, H. (2012). Regional brain wave activity changes associated with fatigue. *Psychophysiology*, *49*(4), 574–582. https://doi.org/10.1111/j.1469-8986.2011.01329.x

De Sanctis, P., Butler, J. S., Malcolm, B. R., & Foxe, J. J. (2014). Recalibration of inhibitory control systems during walking-related dual-task interference: A mobile brain-body imaging (MOBI) study. *NeuroImage*, *94*, 55–64. https://doi.org/10.1016/j.neuroimage.2014.03.016

De Vos, M., Gandras, K., & Debener, S. (2014). Towards a truly mobile auditory brain-computer interface: Exploring the P300 to take away. *International Journal of Psychophysiology*, *91*(1), 46–53. https://doi.org/10.1016/j.ijpsycho.2013.08.010

Debener, S., Emkes, R., De Vos, M., & Bleichner, M. (2015). Unobtrusive ambulatory EEG using a smartphone and flexible printed electrodes around the ear. *Scientific Reports*, *5*(1), 16743. https://doi.org/10.1038/srep16743

Debener, S., Minow, F., Emkes, R., Gandras, K., & de Vos, M. (2012). How about taking a low-cost, small, and wireless EEG for a walk? *Psychophysiology*, *49*(11), 1617–1621. https://doi.org/10.1111/j.1469-8986.2012.01471.x

Dehais, F., Duprès, A., Blum, S., Drougard, N., Scannella, S., Roy, R. N., & Lotte, F. (2019). Monitoring pilot's mental workload using ERPs and spectral power with a six-dry-electrode EEG system in real flight conditions. *Sensors*, *19*(6), 1324. https://doi.org/10.3390/s19061324

Di Flumeri, G., Borghini, G., Aricò, P., Sciaraffa, N., Lanzi, P., Pozzi, S., Vignali, V., Lantieri, C., Bichicchi, A., Simone, A., & Babiloni, F. (2018). EEG-based mental workload neurometric to evaluate the impact of different traffic and road conditions in real driving settings. *Frontiers in Human Neuroscience*, *12*, 509.

Dikker, S., Wan, L., Davidesco, I., Kaggen, L., Oostrik, M., McClintock, J., Rowland, J., Michalareas, G., Van Bavel, J. J., Ding, M., & Poeppel, D. (2017). Brain-to-brain synchrony tracks real-world dynamic group interactions in the classroom. *Current Biology*, *27*(9), 1375–1380. https://doi.org/10.1016/j.cub.2017.04.002

Djebbara, Z., Fich, L. B., Petrini, L., & Gramann, K. (2019). Sensorimotor brain dynamics reflect architectural affordances. *Proceedings of the National Academy of Sciences of the United States of America*, *116*(29), 14769–14778. https://doi.org/10.1073/pnas.1900648116

Donchin, E., & Coles, M. G. (1998). Context updating and the P300. *Behavioral and Brain Sciences*, *21*(1), 152–154. https://doi.org/10.1017/S0140525X98230950

Gabard-Durnam, L. J., Wilkinson, C., Kapur, K., Tager-Flusberg, H., Levin, A. R., & Nelson, C. A. (2019). Longitudinal EEG power in the first postnatal year differentiates autism outcomes. *Nature Communications*, *10*(1), 4188.

Giertuga, K., Zakrzewska, M. Z., Bielecki, M., Racicka-Pawlukiewicz, E., Kossut, M., & Cybulska-Klosowicz, A. (2017). Age-related changes in resting-state eeg activity in attention deficit/hyperactivity disorder: A cross-sectional study. *Frontiers in Human Neuroscience*, *11*, 285. https://doi.org/10.3389/fnhum.2017.00285

Gramann, K., Ferris, D. P., Gwin, J., & Makeig, S. (2014). Imaging natural cognition in action. *International Journal of Psychophysiology*, *91*(1), 22–29. https://doi.org/10.1016/j.ijpsycho.2013.09.003

Gramann, K., Gwin, J. T., Bigdely-Shamlo, N., Ferris, D. P., & Makeig, S. (2010). Visual evoked responses during standing and walking. *Frontiers in Human Neuroscience*, *4*, 202. https://doi.org/10.3389/fnhum.2010.00202

Gramann, K., Gwin, J. T., Ferris, D. P., Oie, K., Jung, T. P., Lin, C. T., De Liao, L., & Makeig, S. (2011). Cognition in action: Imaging brain/body dynamics in mobile humans. *Reviews in the Neurosciences*, *2011*, 593–608. https://doi.org/10.1515/RNS.2011.047

Gratton, G. (1998). Dealing with artifacts: The EOG contamination of the event-related brain potential. *Behavior Research Methods, Instruments, & Computers*, *30*(1), 44–53. https://doi.org/10.3758/BF03209415

Gratton, G., Coles, M. G. H., & Donchin, E. (1983). A new method for off-line removal of ocular artifact. *Electroencephalography and Clinical Neurophysiology*, *55*(4), 468–484. https://doi.org/10.1016/0013-4694(83)90135-9

Gwin, J. T., Gramann, K., Makeig, S., & Ferris, D. P. (2010). Removal of movement artifact from high-density EEG recorded during walking and running. *Journal of Neurophysiology*, *103*(6), 3526–3534. https://doi.org/10.1152/jn.00105.2010

Hinrichs, H., Scholz, M., Baum, A. K., Kam, J. W. Y., Knight, R. T., & Heinze, H. J. (2020). Comparison between a wireless dry electrode EEG system with a conventional wired wet electrode EEG system for clinical applications. *Scientific Reports*, *10*(1), 5218. https://doi.org/10.1038/s41598-020-62154-0

14698986, 2024, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/psyp.14603, Wiley Online Library on [24/08/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Hoffmann, S., & Falkenstein, M. (2008). The correction of eye blink artefacts in the EEG: A comparison of two prominent methods. *PLoS ONE*, *3*(8), e3004. https://doi.org/10.1371/journal.pone.0003004

Hölle, D., Meekes, J., & Bleichner, M. G. (2021). Mobile ear-EEG to study auditory attention in everyday life: Auditory attention in everyday life. *Behavior Research Methods*, *53*(5), 2025–2036. https://doi.org/10.3758/S13428-021-01538-0/FIGURES/5

Jacobsen, N. S. J., Blum, S., Witt, K., & Debener, S. (2021). A walk in the park? Characterizing gait-related artifacts in mobile EEG recordings. *The European Journal of Neuroscience*, *54*(12), 8421–8440. https://doi.org/10.1111/EJN.14965

Jang, K. I., Han, S. Y., Xu, S., Mathewson, K. E., Zhang, Y., Jeong, J. W., Kim, G. T., Webb, R. C., Lee, J. W., Dawidczyk, T. J., Kim, R. H., Song, Y. M., Yeo, W. H., Kim, S., Cheng, H., Rhee, S. I., Chung, J., Kim, B., Chung, H. U., … Rogers, J. A. (2014). Rugged and breathable forms of stretchable electronics with adherent composite substrates for transcutaneous monitoring. *Nature Communications*, *5*(1), 1–10. https://doi.org/10.1038/ncomms5779

Jiang, X., Bian, G. B., & Tian, Z. (2019). Removal of artifacts from EEG signals: A review. *Sensors*, *19*(5), 987.

Jungnickel, E., Gehrke, L., Klug, M., & Gramann, K. (2018). MoBI-mobile brain/body imaging. In *Neuroergonomics: The brain at work and in everyday life* (pp. 59–63). Elsevier. https://doi.org/10.1016/B978-0-12-811926-6.00010-5

Jungnickel, E., & Gramann, K. (2016). Mobile brain/body imaging (MoBI) of physical interaction with dynamically moving objects. *Frontiers in Human Neuroscience*, *10*, 306. https://doi.org/10.3389/fnhum.2016.00306

Klimesch, W., Sauseng, P., & Hanslmayr, S. (2007). EEG alpha oscillations: The inhibition-timing hypothesis. *Brain Research Reviews*, *53*(1), 63–68. https://doi.org/10.1016/j.brainresrev.2006.06.003

Kok, A. (2001). On the utility of P3 amplitude as a measure of processing capacity. *Psychophysiology*, *38*(3), 557–577. https://doi.org/10.1017/S0048577201990559

Krigolson, O. E., Williams, C. C., Norton, A., Hassall, C. D., & Colino, F. L. (2017). Choosing MUSE: Validation of a low-cost, portable EEG system for ERP research. *Frontiers in Neuroscience*, *11*, 109. https://doi.org/10.3389/fnins.2017.00109

Kuziek, J. W. P., Shienh, A., & Mathewson, K. E. (2017). Transitioning EEG experiments away from the laboratory using a raspberry pi 2. *Journal of Neuroscience Methods*, *277*, 75–82. https://doi.org/10.1016/j.jneumeth.2016.11.013

La, T. G., Qiu, S., Scott, D. K., Bakhtiari, R., Kuziek, J. W. P., Mathewson, K. E., Rieger, J., & Chung, H. J. (2018). Two-layered and stretchable e-textile patches for wearable healthcare electronics. *Advanced Healthcare Materials*, *7*(22), e1801033. https://doi.org/10.1002/ADHM.201801033

Ladouce, S., Donaldson, D. I., Dudchenko, P. A., & Ietswaart, M. (2017). Understanding minds in real-world environments: Toward a mobile cognition approach. *Frontiers in Human Neuroscience*, *10*, 694. https://doi.org/10.3389/fnhum.2016.00694

Ladouce, S., Donaldson, D. I., Dudchenko, P. A., & Ietswaart, M. (2019). Mobile EEG identifies the re-allocation of attention during real-world activity. *Scientific Reports*, *9*(1), 1–10. https://doi.org/10.1038/s41598-019-51996-y

Lang, P. J. (1979). A bio-informational theory of emotional imagery. *Psychophysiology*, *16*(6), 495–512.

Lau-Zhu, A., Lau, M. P. H., & McLoughlin, G. (2019). Mobile EEG in research on neurodevelopmental disorders: Opportunities and challenges. *Developmental Cognitive Neuroscience*, *36*, 100635. https://doi.org/10.1016/j.dcn.2019.100635

Leach, S., Chung, K. Y., Tüshaus, L., Huber, R., & Karlen, W. (2020). A protocol for comparing dry and wet EEG electrodes during sleep. *Frontiers in Neuroscience*, *14*, 586. https://doi.org/10.3389/FNINS.2020.00586/BIBTEX

Leal Neto, O., Haenni, S., Phuka, J., Ozella, L., Paolotti, D., Cattuto, C., Robles, D., & Lichand, G. (2021). Combining wearable devices and mobile surveys to study child and youth development in Malawi: Implementation study of a multimodal approach. *JMIR Public Health and Surveillance*, *7*(3), e23154. https://doi.org/10.2196/23154

Leone, C., Feys, P., Moumdjian, L., D'Amico, E., Zappia, M., & Patti, F. (2017). Cognitive-motor dual-task interference: A systematic review of neural correlates. *Neuroscience and Biobehavioral Reviews*, *75*, 348–360. https://doi.org/10.1016/j.neubiorev.2017.01.010

Liebherr, M., Weiland-Breckle, H., Grewe, T., & Schumacher, P. B. (2018). Cognitive performance under motor demands—On the influence of task difficulty and postural control. *Brain Research*, *1684*, 1–8. https://doi.org/10.1016/j.brainres.2018.01.025

Lim, C. K. A., Chia, W. C., & Chin, S. W. (2014). A mobile driver safety system: Analysis of single-channel EEG on drowsiness detection. In *2014 International Conference on Computational Science and Technology (ICCST)* (pp. 1–5). IEEE.

Lopes da Silva, F. (2013). EEG and MEG: Relevance to neuroscience. *Neuron*, *80*(5), 1112–1128. https://doi.org/10.1016/j.neuron.2013.10.017

Lopez-Gordo, M. A., Sanchez-Morillo, D., & Pelayo Valle, F. (2014). Dry EEG electrodes. *Sensors*, *14*(7), 12847–12870. https://doi.org/10.3390/s140712847

Luck, S. J. (2014). *An introduction to the event-related potential technique* (2nd ed.). The MIT Press.

Luck, S. J., & Kappenman, E. S. (2016). Electroencephalography and event-related brain potentials. In J. T. Cacioppo, L. G. Tassinary, & G. G. Berntson (Eds.), *Handbook of psychophysiology* (4th ed.). Oxford University Press. https://doi.org/10.1017/9781107415782.005

Makeig, S., Gramann, K., Jung, T. P., Sejnowski, T. J., & Poizner, H. (2009). Linking brain, mind and behavior. *International Journal of Psychophysiology*, *73*(2), 95–100. https://doi.org/10.1016/j.ijpsycho.2008.11.008

Malcolm, B. R., Foxe, J. J., Butler, J. S., Molholm, S., & De Sanctis, P. (2018). Cognitive load reduces the effects of optic flow on gait and electrocortical dynamics during treadmill walking. *Journal of Neurophysiology*, *120*(5), 2246–2259. https://doi.org/10.1152/jn.00079.2018

Mathewson, K. E., Harrison, T. J. L., & Kizuk, S. A. D. (2017). High and dry? Comparing active dry EEG electrodes to active and passive wet electrodes. *Psychophysiology*, *54*(1), 74–82. https://doi.org/10.1111/psyp.12536

Meiser, A., Tadel, F., Debener, S., & Bleichner, M. G. (2020). The sensitivity of ear-EEG: Evaluating the source-sensor relationship using forward modeling. *Brain Topography*, *33*, 665–676. https://doi.org/10.1007/s10548-020-00793-2

14698986, 2024, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/psyp.14603, Wiley Online Library on [24/08/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Mirkovic, B., Bleichner, M. G., De Vos, M., & Debener, S. (2016). Target speaker detection with concealed EEG around the ear. *Frontiers in Neuroscience*, *10*, 206084.

Mulert, C., Pogarell, O., Juckel, G., Rujescu, D., Giegling, I., Rupp, D., Mavrogiorgou, P., Bussfeld, P., Gallinat, J., Möller, H. J., & Hegerl, U. (2004). The neural basis of the P300 potential: Focus on the time-course of the underlying cortical generators. *European Archives of Psychiatry and Clinical Neuroscience*, *254*, 190–198.

Mullen, T., Kothe, C., Chi, Y. M., Ojeda, A., Kerth, T., Makeig, S., Cauwenberghs, G., & Jung, T. P. (2013). Real-time modeling and 3D visualization of source dynamics and connectivity using wearable EEG. In *In 2013 35th annual international conference of the IEEE engineering in medicine and biology society (EMBC)* (pp. 2184–2187). IEEE.

Nathan, K., & Contreras-Vidal, J. L. (2016). Negligible motion artifacts in scalp electroencephalography (EEG) during treadmill walking. *Frontiers in Human Neuroscience*, *9*, 708. https://doi.org/10.3389/fnhum.2015.00708

Ojeda, A., Bigdely-Shamlo, N., & Makeig, S. (2014). MoBILAB: An open source toolbox for analysis and visualization of mobile brain/body imaging data. *Frontiers in Human Neuroscience*, *8*, 121. https://doi.org/10.3389/fnhum.2014.00121

Olejniczak, P. (2006). Neurophysiologic basis of EEG. *Journal of Clinical Neurophysiology*, *23*(3), 186–189. https://doi.org/10.1097/01.wnp.0000220079.61973.6c

Oliveira, A. S., Schlink, B. R., Hairston, W. D., König, P., & Ferris, D. P. (2016). Proposing metrics for benchmarking novel EEG technologies towards real-world measurements. *Frontiers in Human Neuroscience*, *10*, 188. https://doi.org/10.3389/fnhum.2016.00188

Parasuraman, R., & Rizzo, M. (2007). *Introduction to neuroergonomics* (pp. 3–12). The brain at work. https://doi.org/10.1093/acprof:oso/9780195177619.001.0001

Picton, T. W. (1992). The P300 wave of the human event-related potential. *Journal of Clinical Neurophysiology*, *9*, 456–479. https://doi.org/10.1097/00004691-199210000-00002

Piñeyro Salvidegoitia, M., Jacobsen, N., Bauer, A. K. R., Griffiths, B., Hanslmayr, S., & Debener, S. (2019). Out and about: Subsequent memory effect captured in a natural outdoor environment with smartphone EEG. *Psychophysiology*, *56*(5), e13331. https://doi.org/10.1111/PSYP.13331

Pizzamiglio, S., Naeem, U., Abdalla, H., & Turner, D. L. (2017). Neural correlates of single-and dual-task walking in the real world. *Frontiers in Human Neuroscience*, *11*, 460. https://doi.org/10.3389/FNHUM.2017.00460/BIBTEX

Poldrack, R. A., Kittur, A., Kalar, D., Miller, E., Seppa, C., Gil, Y., Stott Parker, D., Sabb, F. W., & Bilder, R. M. (2011). The cognitive atlas: Toward a knowledge foundation for cognitive neuroscience. *Frontiers in Neuroinformatics*, *5*, 17. https://doi.org/10.3389/fninf.2011.00017

Polich, J., & Comerchero, M. D. (2003). P3a from visual stimuli: Typicality, task, and topography. *Brain Topography*, *15*, 141–152. https://doi.org/10.1023/A:1022637732495

Protzak, J., & Gramann, K. (2018). Investigating established EEG parameter during real-world driving. *Frontiers in Psychology*, *9*, 2289. https://doi.org/10.3389/fpsyg.2018.02289

Protzak, J., Wiczorek, R., & Gramann, K. (2021). Peripheral visual perception during natural overground dual-task walking in older and younger adults. *Neurobiology of Aging*, *98*, 146–159. https://doi.org/10.1016/j.neurobiolaging.2020.10.009

Reiser, J. E., Wascher, E., & Arnau, S. (2019). Recording mobile EEG in an outdoor environment reveals cognitive-motor interference dependent on movement complexity. *Scientific Reports*, *9*(1), 1–14. https://doi.org/10.1038/s41598-019-49503-4

Robles, D., Kuziek, J. W., Lai, J., Mazumder, R., Scanlon, J. E., & Mathewson, K. E. (2022). Surrounding traffic matters: Increases in traffic volume are related to changes in EEG rhythms in urban cyclists. *bioRxiv*, 2022-05. https://doi.org/10.1101/2022.05.27.493782

Robles, D., Kuziek, J. W. P., Wlasitz, N. A., Bartlett, N. T., Hurd, P. L., & Mathewson, K. E. (2021). EEG in motion: Using an oddball task to explore motor interference in active skateboarding. *European Journal of Neuroscience*, *54*(12), 8196–8213. https://doi.org/10.1111/ejn.15163

Rocca, J. (2003). *Galen on the brain: Anatomical knowledge and physiological speculation in the second century AD* (Vol. *26*). Brill.

Scanlon, J. E. M., Jacobsen, N. S. J., Maack, M. C., & Debener, S. (2022). Stepping in time: Alpha-mu and beta oscillations during a walking synchronization task. *NeuroImage*, *253*, 119099. https://doi.org/10.1016/J.NEUROIMAGE.2022.119099

Scanlon, J. E. M., Cormier, D. L., Townsend, K. A., Kuziek, J. W. P., & Mathewson, K. E. (2019). The ecological cocktail party: Measuring brain activity during an auditory oddball task with background noise. *Psychophysiology*, *56*(11), e13435. https://doi.org/10.1111/psyp.13435

Scanlon, J. E. M., Jacobsen, N. S. J., Maack, M. C., & Debener, S. (2021). Does the electrode amplification style matter? A comparison of active and passive EEG system configurations during standing and walking. *The European Journal of Neuroscience*, *54*(12), 8381–8395. https://doi.org/10.1111/EJN.15037

Scanlon, J. E. M., Redman, E. X., Kuziek, J. W. P., & Mathewson, K. E. (2020). A ride in the park: Cycling in different outdoor environments modulates the auditory evoked potentials. *International Journal of Psychophysiology*, *151*, 59–69. https://doi.org/10.1016/j.ijpsycho.2020.02.016

Scanlon, J. E. M., Sieben, A. J., Holyk, K. R., & Mathewson, K. E. (2017). Your brain on bikes: P3, MMN/N2b, and baseline noise while pedaling a stationary bike. *Psychophysiology*, *54*(6), 927–937. https://doi.org/10.1111/psyp.12850

Scanlon, J. E. M., Townsend, K. A., Cormier, D. L., Kuziek, J. W. P., & Mathewson, K. E. (2019). Taking off the training wheels: Measuring auditory P3 during outdoor cycling using an active wet EEG system. *Brain Research*, *1716*, 50–61. https://doi.org/10.1016/j.brainres.2017.12.010

Schmuckler, M. A. (2001). What is ecological validity? A dimensional analysis. *Infancy*, *2*(4), 419–436. https://doi.org/10.1207/S15327078IN0204_02

Seeber, M., Scherer, R., Wagner, J., Solis-Escalante, T., & Müller-Putz, G. R. (2015). High and low gamma EEG oscillations in central sensorimotor areas are conversely modulated during the human gait cycle. *NeuroImage*, *112*, 318–326. https://doi.org/10.1016/J.NEUROIMAGE.2015.03.045

Shipton, H. W. (1975). EEG analysis: A history and a prospectus. *Annual Review of Biophysics and Bioengineering*, *4*(1), 1–13. https://doi.org/10.1146/annurev.bb.04.060175.000245

St. Louis, E., Frey, L., Britton, J., Hopp, J., Korb, P., Koubeissi, M., Lievens, W., & Pestana-Knight, E. (2016). *Electroencephalography*

14698986, 2024, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/psyp.14603, Wiley Online Library on [24/08/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

(EEG): An introductory text and atlas of normal and abnormal findings in adults, children, and infants. American Epilepsy Society. https://doi.org/10.5698/978-0-9979756-0-4

Storzer, L., Butz, M., Hirschmann, J., Abbasi, O., Gratkowski, M., Saupe, D., Schnitzler, A., & Dalal, S. S. (2016). Bicycling and walking are associated with different cortical oscillatory dynamics. *Frontiers in Human Neuroscience*, *10*, 61.

Sur, S., & Sinha, V. K. (2009). Event-related potential: An overview. *Industrial Psychiatry Journal*, *18*(1), 70–73. https://doi.org/10.4103/0972-6748.57865

Vallabhaneni, A., Wang, T., & He, B. (2005). Brain—computer interface. In *Neural engineering* (pp. 85–121). Springer US.

Wagner, J., Solis-Escalante, T., Grieshofer, P., Neuper, C., Müller-Putz, G., & Scherer, R. (2012). Level of participation in robotic-assisted treadmill walking modulates midline sensorimotor EEG rhythms in able-bodied subjects. *NeuroImage*, *63*(3), 1203–1211. https://doi.org/10.1016/J.NEUROIMAGE.2012.08.019

Wascher, E., Heppner, H., Kobald, S. O., Arnau, S., Getzmann, S., & Möckel, T. (2016). Age-sensitive effects of enduring work with alternating cognitive and physical load. A study applying mobile EEG in a real-life working scenario. *Frontiers in Human Neuroscience*, *9*, 711. https://doi.org/10.3389/fnhum.2015.00711

Wascher, E., Rasch, B., Sänger, J., Hoffmann, S., Schneider, D., Rinkenauer, G., Heuer, H., & Gutberlet, I. (2014). Frontal theta activity reflects distinct aspects of mental fatigue. *Biological Psychology*, *96*, 57–65. https://doi.org/10.1016/j.biopsycho.2013.11.010

Wilkinson, C. M., Burrell, J. I., Kuziek, J. W. P., Thirunavukkarasu, S., Buck, B. H., & Mathewson, K. E. (2020). Predicting stroke severity with a 3-min recording from the Muse portable EEG system for rapid diagnosis of stroke. *Scientific Reports*, *10*(1), 1–11. https://doi.org/10.1038/s41598-020-75379-w

Wunderlich, A., & Gramann, K. (2020). Eye-movement related brain potentials during assisted navigation in real-world environments. *European Journal of Neuroscience*, *54*(12), 8336–8354. https://doi.org/10.1111/ejn.15095

Xu, J., & Zhong, B. (2018). Review on portable EEG technology in educational research. *Computers in Human Behavior*, *81*, 340–349. https://doi.org/10.1016/j.chb.2017.12.037

Xu, S., Zhang, Y., Jia, L., Mathewson, K. E., Jang, K. I., Kim, J., Fu, H., Huang, X., Chava, P., Wang, R., Bhole, S., Wang, L., Na, Y. J., Guan, Y., Flavin, M., Han, Z., Huang, Y., & Rogers, J. A. (2014). Soft microfluidic assemblies of sensors, circuits, and radios for the skin. *Science*, *344*(6179), 70–74. https://doi.org/10.1126/science.1250169

Zamrini, E. Y., Meador, K. J., Thompson, W. O., & Lee, G. P. (1991). Reproducibility of p3. *International Journal of Neuroscience*, *61*(1–2), 113–120. https://doi.org/10.3109/00207459108986279

Zhao, C., Zhao, M., Liu, J., & Zheng, C. (2012). Electroencephalogram and electrocardiograph assessment of mental fatigue in a driving simulator. *Accident Analysis and Prevention*, *45*, 83–90. https://doi.org/10.1016/j.aap.2011.11.019

Zink, R., Hunyadi, B., Van Huffel, S., & Vos, M. D. (2016). Mobile EEG on the bike: Disentangling attentional and physical contributions to auditory attention tasks. *Journal of Neural Engineering*, *13*(4), 046017. https://doi.org/10.1088/1741-2560/13/4/046017

How to cite this article: Mathewson, K. E., Kuziek, J. P., Scanlon, J. E. M., & Robles, D. (2024). The moving wave: Applications of the mobile EEG approach to study human attention. *Psychophysiology*, *61*, e14603. https://doi.org/10.1111/psyp.14603

14698986, 2024, 9, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/psyp.14603, Wiley Online Library on [24/08/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License